Lab 15: 60-page packets: section passes first, reconcile second

Topic 1: Agentic Architecture & Orchestration · Task 1.6: Design task decomposition strategies for complex workflows

Runs on Simulator 6: Structured Data Extraction · Run in class

Source: SkillCertPro practice exam Part 1, Q31 (unofficial third-party question)

The question

Production evaluation shows accuracy drops on 60-page packets with predictable sections, such as intake forms, narrative notes, invoices, and attachments. The current single-pass extraction often captures fields within individual sections, but produces duplicate entities, inconsistent totals, and occasional schema-valid output with incorrect relationships. Which task-decomposition strategy should this team use first?

A. Run section-level extraction passes, then run a reconciliation pass that validates schema, deduplicates entities, and checks cross-section consistency.

B. Generate adaptive subtasks after each page, then stop when the agent determines that all required fields have been found.

C. Summarize each packet into a narrative first, then extract the required schema fields from that summary in a final pass.

D. Send the full packet through one expanded extraction prompt, then ask Claude to self-check the JSON before returning it.

Simulate the solution

```
# In the browser: open Simulator 6, then Lab 15
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 15

# In a terminal, from simulators/simulator-6-structured-extraction/
uv run extract.py --case UC-31
```

What you should see

Single pass: one record for three documents. Section pipeline: one record per section.

evidence/cases/UC-31.json shows the single pass returning one record for a packet that holds three distinct documents, and the section pipeline returning four section records reconciled to three unique invoice_ids with the duplicate merged and any total mismatch listed.

Open `simulators/simulator-6-structured-extraction/evidence/` and find it yourself.

The answer

A. Extract per section, then a reconciliation pass checks totals and duplicates.

For predictable workflows, a fixed sequential decomposition beats asking one prompt to do everything: local first, global second. Section passes focus on less evidence and make failures easier to isolate. The reconciliation pass handles constraints that only make sense once every partial extraction exists, such as deduplication, totals, dates and relationships across sections.

Why the others fail

B. Adaptive decomposition suits exploratory work where the next step depends on discoveries. These packets have a known structure.

C. A narrative summary throws away detail before extraction and can hide errors behind valid-looking JSON.

D. The same overloaded single pass. A self-check rarely catches errors in the model's own output.

Check yourself

- Learner states which checks only make sense in the global pass (dedupe, cross-section totals) and why.
- Learner explains why the section pass reduces attention dilution.
- Learner can say when adaptive decomposition would be the better choice instead.

Now change one thing

Change DOC-04's printed total in fixtures/invoices.txt, rerun, and watch by_document_type.invoice fall while the others hold.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.